MILITARY SPECIFICATION SHEET

CHARGER, BATTERY, NICKEL-CADMIUM, AIRCRAFT
30-AMPERE CHARGER, DC INPUT

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the battery charger described herein shall consist of this specification and the latest issue of DOD-C-85050(AS).

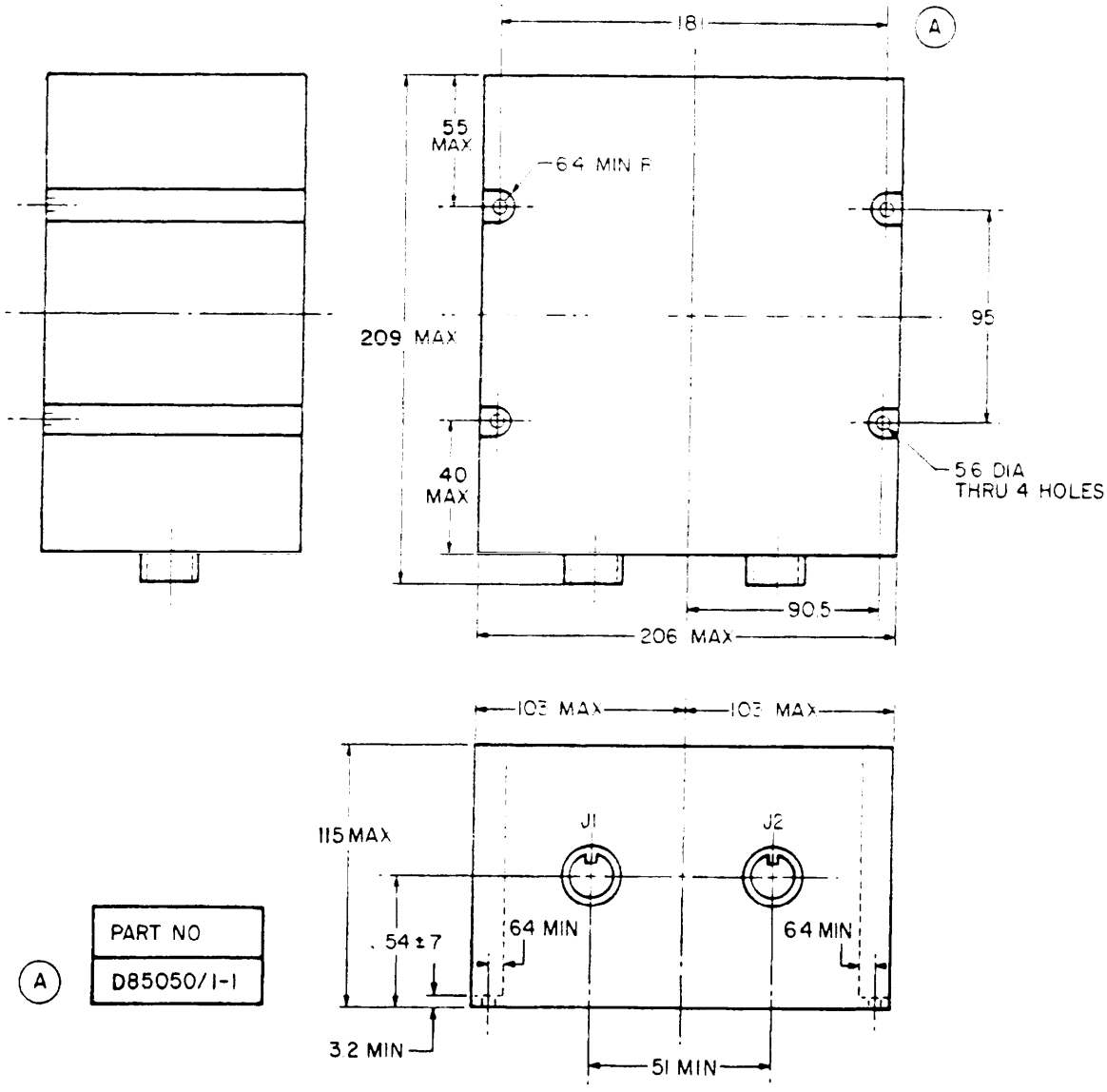


FIGURE 1. Charger outline drawing.

FSC 6130

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REQUIREMENTS:

PART NUMBER: D85050/1-1

DIMENSIONS: See Figure 1. All dimensions on drawing are in millimeters. Unless otherwise specified, tolerance is ±0.4.

WEIGHT: 3.6 Kilograms, maximum.

MOUNTING: In any attitude. See Figure 1 for mounting dimensions.

CONNECTORS:

J1-A	Input Power	J1 = MS3102RX20-22P
J1-B	Spare	
J1-C	Ground	
J1-D	Spare	
J1-E	Battery (+)	
J1-F	Spare	
J2-A	Thermal Switch	J2 = MS3112E14-12P
J2-B	Fault Signal	
J2-C	Spare	
J2-D	Battery Center Tap	
J2-E	Temp Sense	
J2-F	Temp Sense	
J2-G	24V Charge Inhibit	
J2-H	Ground Charge Inhibit	
J2-J, J2-K	Normal Open Relay Contacts for Battery Relay Control	
J2-L, J2-M	30ah Battery Voltage Control	

INPUT POWER: Type I, 28 VDC, 45 amperes maximum steady-state. With zero input voltage applied to pin J1-A, the maximum current drawn by the charger from the battery shall be less than 10.0 milliamperes.

AVERAGE OUTPUT CURRENT TO CHARGE 19-CELL, 30ah BATTERY: (A) 37 amperes maximum.

(A) EFFICIENCY: 82 percent minimum.

BATTERY: Qualified 19-cell, 20ah and 30ah Ni-Cd.

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TEMPERATURE SENSOR: Wirewound, 3000-ohm ±1% at 25°C
T.C. = 3900 ±30 PPM/°C below 25°C
T.C. = 4100 ±30 PPM/°C above 25°C

TEMP	R-OHM (NOMINAL)
-34°C	2305
-29°C	2369
-18°C	2500
25°C	3000
49°C	3294
54°C	3362

MISCELLANEOUS CONTROL FUNCTIONS:

GROUND CHARGE INHIBIT - When Pin H of connector J2 is connected to the aircraft ground, contacts J2-J and J2-K shall close and the charger shall terminate charge. With pin J2-H open, the charger shall operate normally as specified.

24-VOLT CHARGE INHIBIT - When Pin G of connector J2 is connected to 24.0 ± 5.0 VDC, contacts J2-J and J2-K shall close and the charger shall terminate charge. With pin J2-G open, the charger shall operate normally as specified.

30ah BATTERY VOLTAGE CONTROL - When a jumper wire is connected between Pins L and M of connector J2, the mode change voltage which controls the charging mode shall be reduced to be compatible with 30ah batteries. With no jumper wire, the mode change voltage shall be compatible with 20ah batteries.

Preparing activity:
Navy - AS
(Project No. 6130-N239-1)

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